Allianz

Allianz x Bayreuth Hackathon

Konsolidierte Challenge-Übersicht nach Workstream

Workstream: AI Use Cases

1. AI Use Case Prioritization Assistant

Idee / Problemstellung	A tool that helps prioritize AI use cases by business impact in a traceable, defensible way. Rather than prescribing a single scoring mechanism, teams are asked to develop and justify their own decision model, one that can handle incomplete information and conflicting stakeholder views, not just produce a single number.
Problem Statement (Detail)	Allianz business units continuously generate new AI use case ideas, but there is no shared basis to compare them. "Business impact" means something different in each department (cost savings, revenue growth, risk reduction, regulatory necessity), reliable data linking an idea to real financial impact rarely exists before something is built, and decisions are made department by department with no shared format and no systematic check on whether predicted benefits materialized.
Warum relevant	AI investment decisions are currently often made ad hoc, with different implicit criteria depending on the department. This ties up capacity in low-impact projects and delays high-impact ones.
Business-Kontext	Relevant to anyone who makes or prepares AI investment decisions (AI Factory, business unit leadership). Mis-prioritization ties up capacity in low-impact projects and delays high-impact initiatives.
Erwartetes Ergebnis / MVP	Open to the solution space. Possible directions: a multi-criteria decision model that can handle incomplete or uncertain inputs; an approach that makes conflicting stakeholder weightings visible; sensitivity/scenario analysis; optionally an agent that extracts structured evaluation dimensions from unstructured use case descriptions. Expected result is a concept plus a working prototype.
Daten & Tools	Public industry figures (combined ratio, expense ratio) from insurers' annual reports (Allianz, Munich Re, Zurich, AXA) and industry associations (GDV, Insurance Europe); publicly documented AI prioritization frameworks as a starting point; a synthetic use case list as a test set; non-confidential interviews with AI Factory colleagues.
Erfolgskriterien	The model remains traceable/explainable even with more complex methods; it is robust to incomplete or conflicting inputs; it explicitly accounts for the fact that "impact" differs across departments; the result shows genuine methodological work and can be presented understandably to non-experts.
Scope & GenAI-Fit	Sufficient substance for a full semester project. Since the problem is deliberately left open, the team has room to go deeper depending on their

	own strengths (decision theory, uncertainty modeling, NLP-based extraction, or optimization approaches).

2. AI Potential in Reserve Adequacy Assessment

Idee / Problemstellung	Reserving (estimating future claims liabilities) is one of the most consequential and judgment-heavy processes on an insurer's balance sheet, but it is unclear, from a public, outside-in perspective, where AI could meaningfully support this process without compromising actuarial rigor and regulatory scrutiny (Solvency II).
Business-Kontext	Relevant to Actuarial and Finance functions, and to the AI Factory when considering where AI investment in balance sheet adjacent processes could have real leverage. Reserving errors have direct P&L and capital consequences.
Erwartetes Ergebnis / MVP	A concept, grounded in public actuarial and industry literature, of where in the reserving process (e.g. data preparation, pattern detection in claims development, scenario/sensitivity testing, documentation) AI could plausibly add value without replacing actuarial judgment, plus a small demonstrative prototype on synthetic claims data (e.g. a synthetic claims triangle).
Daten & Tools	Public Allianz and peer annual reports (reserve levels, reserving approach disclosures); public actuarial and industry literature on AI/ML in reserving; synthetic claims development data; optionally one orientation conversation with an Actuarial colleague (concept level only).
Erfolgskriterien	The proposal respects actuarial and regulatory constraints rather than treating reserving as a generic prediction problem; identifies a specific, well-scoped point of AI application; the prototype is demonstrable on synthetic data; a non-actuary can understand the problem it addresses.
Scope & GenAI-Fit	Narrow enough to execute well in a semester, since it focuses on one process rather than the whole balance sheet, while leaving room to go deeper into specific reserving methods.

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Workstream: R&D Hub

1. Synthetic Data on Demand

Idee / Problemstellung	A concept and a working tool that takes a short specification (domain, fields, output format) and returns synthetic data matching it. Since nobody knows which data will be needed next, the approach has to work for a domain the team never saw while building it. The tool should generate the data, state how faithful it is, and show what it does not reproduce.
Problem Statement (Detail)	Work on AI in insurance often stalls because the data it needs is restricted. Generating a convincing substitute is harder still to do generically: health records, claims invoices, and policy data each have their own structure and rules of consistency, and there is no agreed way to say how good a synthetic set actually is.
Warum relevant	Teams who want to build or test AI on insurance data often cannot touch real records, because data protection rules and access controls stand in the way. Approval takes time, and hand-built test data is usually too clean and too small to prove anything.
Business-Kontext	Relevant to anyone who builds, tests, or demonstrates AI where real data is off limits (R&D Hub, AI Factory, business units). Teams can start work without waiting for data access and share test sets with partners and universities without disclosing anything.
Erwartetes Ergebnis / MVP	A concept for generating synthetic data across more than one domain, plus a working prototype for at least two contrasting domains (e.g. structured health records and semi-structured claims invoices), with a way to measure fidelity, utility, and privacy risk.
Daten & Tools	Public open datasets as reference for structure and distributions; public literature and open-source libraries on synthetic data generation (generative models, differential privacy); published measures of fidelity/utility/privacy leakage; public regulatory material (GDPR, EU AI Act). No internal or real customer data.
Erfolgskriterien	The approach works on at least two different shapes of data; the team states plainly what the generated data preserves and drops; privacy risk is measured rather than asserted; a downstream task trained on the synthetic set behaves close enough to one using real data.
Scope & GenAI-Fit	Sufficient substance for a full semester, since the team has to build for more than one domain and then prove the output is good. Central role for AI throughout.

2. Digital Twin of a Persona for Health Risk Modelling

Idee / Problemstellung	A digital twin is a running model of a person's health over time, built from public cohort data and fed with the kind of information a health insurer works with (age, sex, height/weight, blood pressure, lab values, activity,

	diagnoses, treatments). The team builds a twin for a synthetic member of a synthetic cohort and uses it to ask what happens if.
Problem Statement (Detail)	Health insurers model risk on broad groups and past claims, which gives little grip on how an individual course develops or on what an intervention is worth over years. There is no widely used way to build such a running model, and the data that would make it easy is exactly the data that cannot be touched.
Warum relevant	Health portfolios are priced and reserved on history and on broad risk groups, which says little about how one member's course develops or what a prevention offer is worth. Published medical risk models predict single endpoints, but do not run forward as a coupled system.
Business-Kontext	Relevant to Health underwriting, actuarial, and product teams, and to anyone weighing where AI investment in health portfolios would have leverage. Prevention and care management programmes are hard to value in advance.
Erwartetes Ergebnis / MVP	A concept plus a working prototype twin for one part of health risk, built on a synthetic cohort and public medical/cost data. The prototype should let a user change something (e.g. treatment adherence, prevention uptake) and see the effect on health path and modelled cost.
Daten & Tools	Public health cohort and survey data (NHANES, national health surveys, WHO/OECD, Eurostat); published clinical risk/progression models and actuarial morbidity tables; public literature on digital twins in medicine; public unit cost data; a synthetic cohort built by the team. No real patient or member data.
Erfolgskriterien	The twin models one part of health risk properly; it runs on a synthetic cohort with no scoring of identifiable people; results are checked against published population outcomes; ethical/legal limits (GDPR special category rules, EU AI Act) are addressed in the design; outputs framed as decision support, never as a decision about a person.
Scope & GenAI-Fit	Sufficient substance for a full semester, since the team has to build the cohort, wire in published risk and cost models, and validate the behaviour against population data.

3. Pricing in Commercial Insurance

Idee / Problemstellung	Commercial pricing works differently from retail: few comparable risks, single accounts can be very large, much information arrives as documents, and underwriters' judgement carries the decision. Teams are asked to find where AI genuinely helps, e.g. pulling structure out of submission documents, finding comparable risks, or flagging where a quoted price sits far from the pattern, keeping the underwriter in the decision.
Problem Statement (Detail)	Pricing in commercial insurance rests on sparse data and expert judgement. Each account is close to unique, exposure information arrives as text and spreadsheets, and there are too few comparable risks to fit the kind of model that works in retail lines.
Warum relevant	Retail lines have plenty of data and mature models. Commercial lines have thin data per account and rich information locked in documents, so the

	same techniques do not carry over. Getting a price wrong on a large account has a direct effect on the result.
Business-Kontext	Relevant to Underwriting, Actuarial, and anyone considering where AI investment in commercial lines would have leverage. Pricing errors on large accounts feed straight through to the result.
Erwartetes Ergebnis / MVP	A concept for where AI supports commercial pricing, grounded in public literature and market data, plus a prototype for one well-chosen part on synthetic data (e.g. extracting structured exposure information, finding comparable risks, flagging outlier quotes). (Any output is a draft for human review, never a final price.)
Daten & Tools	Public commercial lines statistics and market reports (GDV, Insurance Europe, Lloyd's, rating agencies); public annual reports with commercial/speciality segment disclosures; public actuarial literature on pricing sparse, heterogeneous risks; synthetic submission documents and portfolio; public regulatory material on EU AI Act.
Erfolgskriterien	The work respects what makes commercial pricing different rather than treating it as a large regression problem; the underwriter stays in the decision; the prototype runs on synthetic data and shows its reasoning; regulatory limits addressed in the design.
Scope & GenAI-Fit	Sufficient substance for a full semester, since the team has to understand the trade before it can model it, then pick one part to build well.

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4. Rebuild with AI: Understand an App, Recreate It on a Modern Stack

Idee / Problemstellung	An AI tool that looks at an existing app (its code, docs, and anything written about it) and works out what it actually does. It writes this up twice: once in plain business language, and once as a technical deep-dive linking back to the real code. Business people and engineers then correct it together. Once everyone agrees on what the new version should do, the same tool builds it and checks it works as agreed. The app does not have to be old — a team can pick any app and rebuild it with new ideas on a modern stack.
Problem Statement (Detail)	Before a team can safely rebuild a system, someone has to work out what it actually does: what are its business features, how is it built, where does documentation disagree with the real thing. Today this is done by hand, slowly, producing one report that is too technical for business people and too shallow for engineers — and hard to correct once building has started.
Warum relevant	Rebuilding an app is rarely slow because writing new code is hard — it is slow because nobody fully understands the old app anymore. Documentation goes out of date, and business people and engineers often describe the same app differently, so teams redo work later because they agreed on the wrong plan too early.
Business-Kontext	Relevant to anyone who runs or decides on rebuild projects: an AI Factory rebuild team, enterprise architects, or an app owner deciding whether to rebuild, replace or shut down a system. Most rebuild projects go wrong at the understanding stage.
Erwartetes Ergebnis / MVP	A working first version of an AI tool with three steps: Understand (read a chosen app's code and docs, write it up as a business story plus a technical deep-dive, each claim linked to the real file/function/document); Reshape (let a business person and an engineer correct the write-up until both agree); Build (turn the agreed plan into a real, working version of an important part of the app, checked automatically against what was agreed).
Daten & Tools	A real app the team picks itself (old or one to redo with new ideas), with its own public documentation, README, issue tracker and commit history; public articles on rebuilding and migrating old systems (e.g. strangler-fig pattern); short, non-confidential conversations with AI Factory people to sanity-check the idea; no access to internal systems or data.
Erfolgskriterien	The business story and technical deep-dive come from the same source; every claim links back to something real; a business person can read and correct the story without reading code; when someone corrects something in Reshape, the plan actually changes; the final build is checked against agreed behaviour; the whole process is shown live on two or three apps the team picked itself.
Scope & GenAI-Fit	During the opening weekend, get one chosen app through the whole process on a small part of it. Use the rest of the semester to try more apps, go deeper on each step, and improve the pictures and mock-ups. Stretch goal: add a rough cost/time estimate and short business case from the same write-up.

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Workstream: Personal Productivity

1. Shareable Knowledgespace Across the Team and AI Tools ("Second Brain")

Idee / Problemstellung	A shared, tool-agnostic "second brain" that a team builds up together and that any AI assistant (Claude, Copilot, ChatGPT, or whichever tool a person prefers) can read from and write to. The knowledge itself lives in one common space; the assistant is just the interface. The system also classifies each piece of knowledge by who it matters to (team, department, or tribe), routing it into the right tier automatically, and lets the team mark certain knowledge as restricted.
Problem Statement (Detail)	People on the same team increasingly use different AI assistants, each keeping its own separate context and history, so the same ground gets covered twice. On top of that, nothing distinguishes what only matters to one team from what an entire department should know, so broadly useful knowledge stays buried and sensitive knowledge can be exposed further than it needs to be.
Warum relevant	Teams increasingly use several different AI assistants side by side, and each one starts from zero — notes, decisions and research captured through one tool are invisible to a colleague using another. Today's knowledge also sits in one flat pile, so broadly useful knowledge stays buried while sensitive content may be over-exposed.
Business-Kontext	Relevant to any organization piloting more than one AI assistant at once, and to any organization structured into teams nested inside larger departments. Without a shared layer, AI investment sits in per-tool silos.
Erwartetes Ergebnis / MVP	A common knowledge store that the team builds live, queryable by multiple AI assistants through a standard interface (e.g. an MCP server); a relevance classifier routing new knowledge into the right storage tier; a permissioning layer letting a contributor mark an item as restricted, enforced at retrieval time. Expected result: a working prototype where at least two AI assistants read from and add to the same live knowledge space, with visible tiers and at least one working access restriction.
Daten & Tools	The team's own real notes, decisions and research, captured live during the hackathon — no synthetic dataset needed. Reference: public documentation on Model Context Protocol (MCP) and connector ecosystems; public material on knowledge-management/"second brain" methods (Zettelkasten, PKM) and retrieval/vector-store patterns; public material on text classification and role-based access control (RBAC).
Erfolgskriterien	Knowledge added through one assistant is retrievable and useful through a different assistant; the relevance classifier consistently routes team-only versus broader-audience knowledge, and a person can correct a wrong classification; restricted knowledge is never retrievable by someone outside the defined group, even indirectly; the space keeps working without any single AI tool owning it.

Scope & GenAI-Fit	Because the team builds real content live, visible progress is possible even within a short hackathon timeframe, while the relevance-classification and access-control layer gives room to go deeper over a full semester.

Workstream: Ambassadors

1. AI Project Security Manager

As speed of AI Adoption increases, project teams need to be able to identify key security risks and make decisions around them earlier and throughout the AI project lifecycle. Today multiple security related assessments exist, some of them are still manual which slow down the risk related decisions and deliverables**. Quantifizierter Impact:** Up to 60%+ reduction in time/effort required for AI security review/risks assessments per AI project, and an estimated 50% reduction in lead time required for AI risk management decisions. Assuming 50 AI projects annually, this could save approx. 300-750 security engineering hours per year**. Business Value:** Creates a scalable security capability for assisting AI projects that reduces manual effort and skillset gaps while accelerating risk based decisions earlier in the project lifecycle. Improves cyber resilience and compliance of adopted AI enabled services. **Zielergebnis / MVP** The challenge is to build a working MVP of an AI Project Security Manager that guides an AI project team through a simplified project security lifecycle with 4 defined security gates: (a) AI solution scope & design security review, (b) risks identification & impact assessment, (c) decision for risk prioritization & mitigation and (d) effectiveness validation based on post implementation evidence. The MVP solution should evaluate available project documents, identify missing information, identify AI specific risks, assess their possible impact, prioritize the risks, recommend security controls to neutralize the most severe risks, review decisions and validate evidence from previous steps and provide clear GO /GO WITH CONDITIONS / NO GO decision recommendations with supporting rationale that project teams can use to determine the appropriate next steps. **Success Criteria MVP:** The MVP demonstrates an end to end security lifecycle workflow with defined security gates included, traceable decisions, identified and prioritized risks and corresponding safeguards/controls and their post implementation validation based on provided inputs as security related deliverables for the given AI project scope.. **Benötigte Skills**: Python, AI/LLM security basics, prompt engineering, basic of IS or cybersecurity, data analysis, potentially frontend/UI development. Combines AI/LLM Capabilities with cybersecurity decision support. The innovation is an AI Security Project Manager that evaluates risks, validates previously taken decisions, and implemented controls, determines whether they can proceed through defined security gates. **Team Fit**: Yes, a team of 5 students can deliver a functional MVP within 4 months. **Benötigte Allianz-Daten:** No **Mentor-Verfügbarkeit:** Yes